Xavier Wong

Email: xavierwongzh@gmail.com | LinkedIn | Portfolio

SUMMARY

Engineering Systems and Design graduate specialising in operations research and business analytics, with an AI minor. Experienced in building optimisation models, BI dashboards, and automated data pipelines, including a Power BI solution that saved $2M in SAP enhancement costs. Skilled in Python, SQL, Power BI, and Pyomo.

WORK EXPERIENCE

Razer (Asia-Pacific) Pte. Ltd — **Singapore**
Data Analyst Intern — **Aug 24 – Jan 25**

- Looped into a cross-team discussion, proposed a Power BI alternative to a SAP feature enhancement, saving the company **$2M**.
- Built and automated **data pipelines** using Power Automate to handle SFTP uploads, SharePoint report distribution, and scheduled email reports, improving data transfer efficiency by **25%**.
- Built and maintained **12+ Power BI dashboards** across revenue, marketing, and product views, including a staggered refresh workaround during Black Friday for near real-time monitoring under Power BI Pro constraints.
- Kickstarted **ML adoption** in the ecommerce team through RFM segmentation to identify high and low value customers.

Republic of Singapore Navy — **Singapore**
Robotic Process Automation Specialist — **Jan 21 – Sep 22**

- Built **automation bots** using UiPath to eliminate repetitive admin tasks such as batch data entry and form filling, significantly reducing manual workload for the department.
- Developed a **Telegram bot** to streamline internal communications between units and the personnel hub, removing the need for walk-ins and phone calls for routine queries.

EDUCATION

Singapore University of Technology and Design — **Singapore**
Bachelor of Engineering (Honours) — **Sep 22 – May 26**

- **Major:** Engineering Systems and Design
- **Specialisations:** Business Analytics & Operations Research
- **Minor:** Artificial Intelligence

Temasek Polytechnic — **Singapore**
Diploma in Mechatronics Engineering with Merit — **Apr 17 – Apr 20**

PROJECTS

DUBY (Duty Buddy) | *Capstone Project, St Andrew's Community Hospital*

- Built a **MILP optimisation model** in Pyomo to auto-generate compliant, fair nurse rosters, with CP-SAT as a fallback solver to ensure the system always produces a usable output.
- Piloted at Ward 6, SACH, reducing rostering time from over 3 hours to under 15 minutes per cycle, a **91.7% reduction**, with the algorithm receiving commendation from both nurse managers.

Automated Trading Analytics System | *Personal Project*

- Built a fully automated paper trading system combining LLM-driven decision-making with an evidence-based options flow scoring engine (order composition, open interest deltas, IV drift) to generate directional signals.
- Designed a risk management layer with automated stop-loss/take-profit brackets, drawdown circuit breakers, and position protection checks, plus an evaluation module benchmarking performance against QQQ/SPY.
- Deployed with Python, GitHub Actions scheduling, Supabase, and a Next.js dashboard for monitoring trades and decision logs.

San Diego Airport Simulation | *Airport Systems Modelling & Simulation*

- Processed real flight movement data from **Flightradar24**, cleaned and structured in R, to generate demand scenarios for discrete-event simulation in CAST across multiple operational strategies
- Identified that **ground layout constraints**, were the primary bottleneck at one of the world's busiest single-runway airports, challenging the theoretical assumption that reduced wake turbulence separations would meaningfully improve throughput

TECHNICAL SKILLS

- **Programming & Data Tools:** Python, SQL, R, DAX, Power Query
- **Data Analysis & Machine Learning:** Statistical Analysis, Machine Learning, Forecasting
- **Data Visualization & BI:** Power BI
- **Optimization & Operations Research:** Pyomo, Gurobi, LP/MILP Modeling
- **Process Automation:** Power Automate, UiPath, Google Apps Script